

University College Cork

School of Engineering

EE6019 Research Project - Final Report

A Patient-Specific Seizure Detection Framework for Scalp EEG: Event- Level Evaluation, Feature Reduction, and Deployment-Oriented Analysis

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Declaration

This report was written entirely by the author, except where stated otherwise. The source of any material not created by the author has been clearly referenced. The work described in this report was conducted by the author, except where stated otherwise.

Signed: Yangdeyi Yang Date: 17/04/2026

Abstract

Automated seizure detection from scalp electroencephalography (EEG) remains a difficult research problem. Seizure activity is rare, varies across patients, and is often mixed with noisy and changing background signals. In practice, a useful detector therefore needs high event-level sensitivity, a low false alarm burden, and short detection delay rather than strong window-level accuracy alone. This report presents the design, implementation, and evaluation of a patient-specific seizure detection framework based on the CHB-MIT scalp EEG database.

The proposed framework builds on the classical feature-based approach of Shoeb and Gutttag and extends it in several practical ways. It uses a richer spectral-spatial-temporal representation based on 22 bipolar EEG channels, 20 spectral energy bins per channel, three inter-hemispheric synchrony features, and four-epoch temporal stacking, giving a 1772-dimensional input vector. It also compares three model families, namely SVM with RBF kernel, Random Forest, and XGBoost, under a common patient-specific leave-one-seizure-out (LOSO) protocol with event-level evaluation. A two-stage search procedure is then used to select both the feature-extraction settings and the final Random Forest configuration.

The selected final model is a patient-specific Random Forest using the top 30 features, 500 trees, maximum depth 12, Butterworth SOS filtering, and three previous epochs of temporal history. On the 10-patient subset used in the final implementation, LOSO validation gives a macro sensitivity of 0.98, a median false alarm rate of 0.2455 per hour, and a mean detection delay of 10.64 s, equivalent to about 5.89 false alarms per 24 h. In pooled event terms, the final model detects 53 of 55 seizures across 580.57 h of EEG. Compared with the full-feature Random Forest, the top-30 version improves event sensitivity slightly, reduces the false alarm burden, and makes the model easier to interpret.

The main contribution of this work is not the introduction of a new classifier, but the development of a more complete and deployment-oriented seizure detection pipeline. Compared with the original Shoeb baseline, this project uses a richer feature space, compares several model families under a common event-level LOSO protocol, and carries the final model through feature selection, interpretation, export, and runtime analysis. An additional lightweight branch further examined model compression through a validated 120-tree RF comparison, while a separate smallest-node single-patient compact export example was used only to demonstrate embedded compression feasibility. It therefore provides a stronger and more practical patient-specific benchmark for future work on clinically useful and deployment-aware seizure detection.

Index Terms— seizure detection, scalp EEG, patient-specific modelling, Random Forest, leave-one-seizure-out (LOSO), event-level evaluation, model compression, deployment-oriented framework.

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Glossary of Terms

Table 1. Abbreviations used in this report

Abbreviation	Meaning
EEG	Electroencephalography
EDF	European Data Format
PSD	Power Spectral Density
LOSO	Leave-One-Seizure-Out
SVM	Support Vector Machine
RF	Random Forest
XGB	XGBoost
FAR	False Alarm Rate
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
PR	Precision-Recall
AP	Average Precision
CNN	Convolutional Neural Network
MCU	Microcontroller Unit
RMS	Root Mean Square
TP / FP / FN / TN	True Positive / False Positive / False Negative / True Negative

Symbols and SI Units

Table 2. Symbols and units used in the report

Symbol	Meaning	Unit
f_s	EEG sampling frequency	Hz
T_e	Epoch length	s
h	Number of historical epochs stacked before the current epoch	epochs
K	Number of selected top features	features
x	Feature vector for one EEG epoch or stacked epoch context	-
τ	Decision threshold applied to the predicted probability	-
N_{det}	Number of detected seizure events	events
N_{true}	Number of true seizure events	events
FAR/h	False alarm rate per hour	alarms / hour
d	Detection delay from clinical onset to first alarm	s

1. Introduction

1.1 Background and Motivation

Epileptic seizures are brief episodes of abnormal electrical activity in the brain. Long-term electroencephalography (EEG) monitoring plays an important role in diagnosis and clinical management, but reviewing many hours of multichannel EEG manually is slow, demanding, and difficult to scale. This creates a clear need for automated methods that can assist clinicians by flagging suspicious EEG segments for further review [1], [2], [20].

However, automated seizure detection from scalp EEG remains difficult. Seizure activity occupies only a small proportion of most recordings, the signal is often affected by motion and other artefacts, and seizure patterns vary substantially across patients. These factors make the problem both clinically important and technically challenging [1], [2], [20].

They also make it easy to report results that appear strong in theory but are less useful in practice. In a highly imbalanced dataset, for example, a model that predicts non-seizure for almost every epoch may still achieve high classification accuracy while failing to detect the events that matter clinically. For this reason, the present work focuses primarily on event-level sensitivity, false alarm burden, and detection delay rather than raw window-level accuracy alone [4].

1.2 Problem Statement

The core problem addressed in this project is the development of a patient-specific seizure detection framework for scalp EEG that is technically rigorous, clinically interpretable, and sufficiently deployment-oriented to support future embedded implementation. More specifically, the framework should satisfy four requirements:

1. it should use a validation protocol that limits optimistic bias and reduces the risk of data leakage;
2. it should support interpretation at both the feature level and the event level;
3. it should remain compact enough to motivate later deployment-focused work;
4. it should distinguish clearly between retrospective validation evidence and real-world deployment readiness.

The project builds on the classical feature-based approach introduced by Shoeb and Guttag, but extends it into a broader engineering study. Rather than testing a single classifier in isolation, the work develops a complete pipeline for feature extraction, model comparison, threshold selection, feature reduction, patient-level export, and clinical error analysis under a common patient-specific evaluation framework.

1.3 Project Objectives

The main objectives of the project are:

1. to implement a robust patient-specific seizure detection pipeline for the CHB-MIT scalp EEG database;
2. to design an interpretable spectral-spatial-temporal feature representation based on bipolar scalp EEG channels;
3. to compare SVM with RBF kernel, Random Forest, and XGBoost under a common leave-one-seizure-out (LOSO) event-level validation protocol;
4. to select and refine a final model using leakage-controlled threshold selection and top-k feature reduction;
5. to assess the selected system in terms of detection performance, interpretability, computational profile, and suitability for future deployment-oriented work.

A further objective is to identify the main remaining weaknesses of the system, especially delayed detections and false positives, and to use these observations to define a clear direction for follow-on thesis research.

1.4 Main Contributions and Novelty

The novelty of this work does not lie in proposing a new machine learning algorithm. Instead, it lies in the way the seizure detection problem is structured, evaluated, and carried closer to practical use. The main contributions of the project are as follows:

1. the development of a richer feature representation than the classical Shoeb baseline, combining 22 bipolar channels, 20 spectral bins per channel, spatial synchrony features, and short-term temporal context;
2. a fair benchmark across three model families, namely SVM-RBF, Random Forest, and XGBoost, under a common patient-specific LOSO protocol with event-level evaluation;
3. a two-stage model selection process that first identifies the feature-extraction settings and then selects the final Random Forest configuration and top-k feature subset;
4. the reduction of the final model to a compact top-30 feature representation that improves interpretability and supports deployment-oriented analysis;
5. the integration of patient-level model export, feature-importance analysis, clinical visualisation, runtime profiling, and structured error analysis within one coherent workflow.
6. an exploratory lightweight branch that combines validated RF compression analysis with a separate compact export feasibility example.

Taken together, these contributions move the project beyond a simple notebook demonstration and toward a more complete research prototype. The result is not only a seizure detector, but also a clearer benchmark and a stronger basis for future work on clinically meaningful and deployment-aware EEG analysis.

1.5 Report Structure

Chapter 2 reviews the relevant literature and places the project within both the classical feature-based seizure detection literature and more recent deep-learning work. Chapter 3 introduces the background theory underlying the feature representation, model choices, and evaluation metrics. Chapter 4 describes the dataset, software environment, and experimental framework. Chapter 5 presents the proposed patient-specific seizure detection framework and its implementation. Chapter 6 reports the experimental results and discusses their clinical and technical significance. Chapter 7 summarises the main findings, reflects on the limitations of the study, and states the overall conclusions. Chapter 8 outlines how the present work can be extended in the follow-on summer research module and developed into a fuller thesis.

2. Literature Review and State of the Art

2.1 Why Seizure Detection from EEG Is Difficult

Automated seizure detection from EEG remains a difficult problem because the signal is high-dimensional, non-stationary, patient-dependent, and often affected by noise and artefacts [1], [2], [20]. In long-term recordings, seizure activity typically occupies only a small fraction of the total recording time, which creates severe class imbalance. In addition, the boundary between ictal and non-ictal EEG is not always clear. Rhythmic artefacts, transitional activity, and some non-epileptic patterns may resemble seizures, especially when decisions must be made from short EEG segments [2].

Recent review work confirms that seizure detection performance is strongly influenced not only by model choice, but also by dataset characteristics, preprocessing strategy, feature design, evaluation protocol, and whether the study is patient-specific or patient-independent [2]. This point is important because strong reported performance can sometimes reflect a favourable protocol rather than a detector that would remain reliable in realistic long-term monitoring. A useful seizure detector must therefore be judged by clinically meaningful behaviour, not by classification accuracy alone.

2.2 Patient-Specific Versus Patient-Independent Detection

A central design choice in EEG seizure detection is whether to build one generic model for all patients or a separate model for each patient. Patient-independent systems are attractive because they aim to generalise to unseen subjects. However, they must handle large inter-patient variability in seizure morphology, background EEG, montage conditions, and noise characteristics. In practice, this makes generalisation difficult, especially in scalp EEG, where seizure signatures may be subtle and less spatially focal than in intracranial recordings [2], [17].

By contrast, patient-specific systems exploit the fact that seizure patterns within one patient are often more consistent than seizure patterns across a cohort. This can improve sensitivity and reduce false alarms when retrospective data are available for the same patient. Shoeb and Guttag established this line of work clearly in scalp EEG, and more recent patient-specific studies continue to show that personalised models remain competitive, especially when

clinical interpretability, operating-point control, or wearable deployment are important considerations [1], [18].

Patient-independent seizure detection remains an important goal. For example, Yang et al. investigated patient-independent seizure event detection on the CHB-MIT scalp EEG dataset and showed that this setting is feasible but challenging, precisely because the model must absorb broader cross-subject variability [17]. For the present project, the patient-specific route is therefore a deliberate design choice rather than a limitation hidden after the fact. It reflects the project's aim of building a strong and interpretable benchmark under a realistic personalised setting before moving to more difficult cross-subject generalisation.

2.3 The Shoeb Baseline and Its Continuing Importance

Shoeb and Gutttag remain a key reference point in automated scalp EEG seizure detection [1]. Their work is methodologically important for three reasons. First, it demonstrated that seizure detection could be framed as a structured machine learning problem built on spectral, spatial, and temporal EEG features. Second, it reported clinically meaningful performance measures, including sensitivity, false alarms per 24 h, and latency, rather than relying on raw epoch-level accuracy alone. Third, and most importantly, it warned that random splitting of neighbouring EEG segments can produce misleadingly optimistic results because adjacent windows are highly correlated [1].

That warning remains highly relevant. In their study, performance was measured with record-level separation rather than random epoch-level splitting, and this methodological point still shapes good practice in the field today [1]. The present project therefore treats Shoeb and Gutttag not simply as a historical baseline, but as a methodological anchor. At the same time, it is important to note that direct numerical comparison across studies must always be made carefully, because datasets, patient subsets, annotation choices, and evaluation pipelines are rarely identical.

Figure 1 is reproduced from Shoeb and Gutttag [1] and illustrates why seizure detection is still well suited to patient-specific modelling. The seizure pattern evolves over time, is distributed across multiple channels, and does not reduce to one simple waveform template. This is one reason why structured feature-based approaches remain valuable even in the era of deep learning.

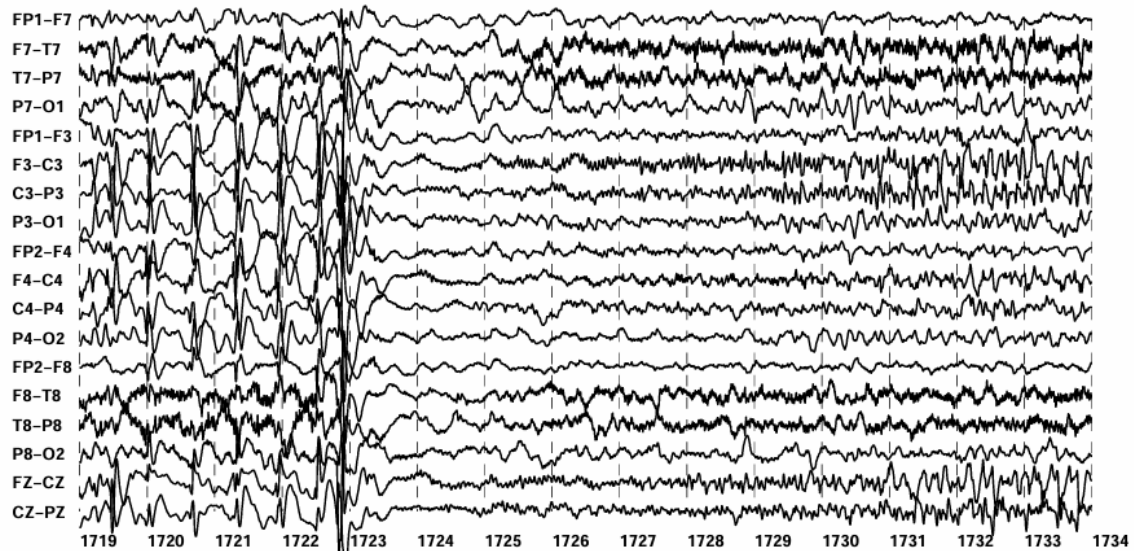


Fig. 1. Representative scalp EEG seizure segment reproduced from Shoeb and Gutttag [1].

2.4 Performance Assessment and the Importance of Event-Level Metrics

How seizure detection performance is measured is just as important as how the detector is built. Temko et al. argued that detector performance should not be reduced to a single metric or to segment-level accuracy alone [4]. Although their study addressed neonatal seizure detection rather than scalp epilepsy monitoring, their methodological argument remains highly relevant here: event-based and segment-based evaluations answer different questions, and both the reporting protocol and the metric definition must be made explicit [4].

This is especially important in seizure detection because the clinical meaning of an error depends on its form. A detector that labels isolated short windows correctly may still be poor if it misses seizure events, generates too many false alarms, or raises alarms too late to be useful. For that reason, event sensitivity, false alarm rate per hour, and detection delay are widely regarded as more informative than accuracy alone when discussing clinical usefulness [1], [4], [18].

The literature also shows that reported performance is often difficult to compare across studies because false alarm definitions, post-processing rules, channel selections, and data partitions vary substantially [2]. This reinforces the need for a clear and internally consistent evaluation protocol. In the present work, event-level performance is treated as primary because it aligns more closely with how a clinician would experience a detector during long-term monitoring.

2.5 Classical Feature-Based Systems Versus End-to-End Deep Models

Classical feature-based machine learning still has several strengths in seizure detection. It offers direct control over feature design, makes it easier to interpret which channels and frequency ranges influence decisions, and often has a clearer engineering path when computational resources are limited. These advantages remain important in patient-specific and deployment-oriented settings, where model transparency and controllable complexity matter as much as raw predictive power [1], [2].

At the same time, deep learning has become increasingly prominent in seizure detection research. Recent reviews describe the widespread use of CNNs, recurrent models, hybrid CNN-RNN systems, attention-based models, and transfer learning pipelines for EEG seizure analysis [2], [19], [20]. These methods can learn richer representations directly from waveform or time-frequency inputs and have produced strong reported performance in many studies. However, they also introduce important trade-offs. They may require more training data, involve higher computational cost, and provide less transparent decision logic than feature-based systems [2], [19].

More recent scalp EEG studies also show that deep learning is not only being used for full-montage hospital recordings, but increasingly for reduced-channel and wearable-oriented seizure detection. For example, Chung et al. proposed a patient-specific deep learning approach on the CHB-MIT dataset and reported that reduced-channel and even single-channel detectors could still achieve strong event-level performance when channel selection was guided by clinically relevant seizure locations [18]. Likewise, recent wearable feasibility studies suggest that reduced-channel seizure detection is clinically attractive, but they also underline the practical importance of comfort, robustness, and real-time usability, not just laboratory performance [21].

For the present project, this literature does not imply that feature-based methods are outdated. Rather, it suggests that a strong, interpretable, and event-level benchmark remains valuable. Any later deep learning model should be judged against such a benchmark under a comparable protocol, instead of being compared only against weaker baselines or accuracy-based results.

2.6 State-of-the-Art Trends Relevant to This Project

Across the recent literature, several trends are especially relevant to the present study. First, evaluation protocol design remains a major determinant of reported performance. Second, patient-specific modelling continues to be competitive, particularly in scalp EEG and limited-data settings. Third, event-level reporting and false alarm analysis are essential for clinically meaningful interpretation. Fourth, temporal context remains important, whether it is encoded explicitly through engineered features or learned implicitly by deep sequential architectures. Finally, interpretability, computational efficiency, and deployment realism remain open challenges even when raw classification performance appears strong [1], [2], [18], [20], [21].

These trends support the design logic of the present project. The goal here is not to claim the most complex classifier, but to build a patient-specific detection framework that is methodologically careful, clinically interpretable, and sufficiently deployment-aware to support further work.

2.7 Research Gap and Positioning of the Present Work

The literature contains strong classical studies, many recent deep learning approaches, and an increasing number of reduced-channel or wearable-oriented seizure detection papers. However, a clear gap remains between these strands of work. Classical feature-based studies often provide interpretability and structured evaluation, but may not always carry the final

model toward deployment-oriented analysis. Recent deep learning studies often report high performance, yet many rely on heterogeneous protocols, limited interpretability, or segment-level metrics that make clinical comparison difficult [2], [18], [19], [20].

There therefore remains a need for a compact and interpretable patient-specific scalp EEG seizure detection framework that is benchmarked fairly, evaluated primarily at the event level, and carried far enough forward to support feature reduction, model export, runtime profiling, and structured clinical error analysis. This is the space that the present project aims to occupy.

In that sense, the contribution of the current work is not the invention of a new classifier, but the construction of a more complete research workflow around seizure detection: one that combines richer but still interpretable features, controlled patient-specific validation, clear event-level assessment, and a stronger bridge from retrospective benchmarking to future deployment-oriented development.

3. Methodological Foundations

3.1 Epoching and Label Assignment

EEG is a continuous time series, but most machine learning models require input samples of fixed length. A common approach is therefore to divide the continuous recording into short epochs and assign one label to each epoch. This converts a long multichannel signal into a sequence of structured samples that can be processed consistently across recordings.

In seizure detection, epoch labeling must be handled carefully because seizure onset and termination rarely align exactly with epoch boundaries. For this reason, an epoch is typically labelled as seizure-positive if it overlaps an annotated seizure interval. This is a practical choice because it reduces the risk of discarding epochs that contain clinically relevant seizure activity near the beginning or end of an event.

At the same time, epoch labels should not be confused with the final clinical target. The clinically meaningful question is not whether each short window is classified correctly in isolation, but whether the system detects seizure events with an acceptable false alarm burden and delay. Epoch-level labeling is therefore best understood as an intermediate representation that supports later event-level analysis rather than as the final objective of the detection system.

3.2 Spectral, Spatial, and Temporal Feature Representation

3.2.1 Spectral Representation

Seizure activity often produces characteristic changes in EEG frequency content. For this reason, frequency-domain features remain a common and effective choice in seizure detection. Instead of representing each epoch only in the time domain, it is often more informative to describe how signal energy is distributed across frequency ranges. This provides a compact summary of rhythmic activity and can reveal differences between seizure and non-seizure patterns that are less obvious in raw waveforms.

A broad-band representation such as delta, theta, alpha, and beta is useful, but a finer frequency partition can preserve more detail. In the present project, spectral energy is represented using narrow frequency bins rather than only a few canonical bands. This choice supports two goals. First, it provides a richer description of seizure-related frequency structure. Second, it improves interpretability, because each selected feature can later be traced to a specific channel, frequency interval, and temporal position.

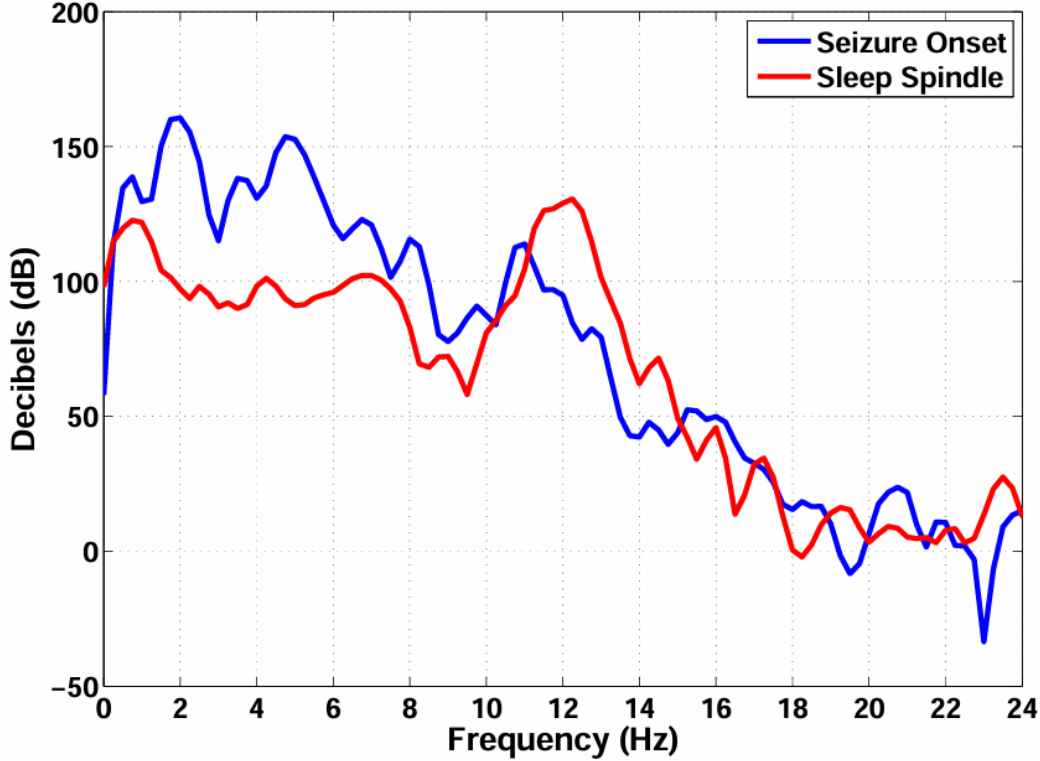


Fig. 2. Spectral contrast between seizure onset and a non-ictal sleep spindle reproduced from Shoeb and Guttag [1].

3.2.2 Spatial Representation

Spectral power alone does not fully capture how seizure activity is distributed across the scalp. In multichannel EEG, spatial relationships between channels can also carry useful information. This is especially relevant when homologous regions in the left and right hemispheres show coordinated or asymmetric activity.

One simple way to include spatial information is to compute synchrony measures between selected channel pairs. In this project, inter-channel correlation is used to provide a compact description of spatial coordination. Although such features are much fewer in number than the spectral features, they add an important extra dimension to the representation. They allow the model to consider not only how much activity exists in a channel, but also how activity is related across regions.

3.2.3 Temporal Representation

A seizure is a dynamic process rather than a static pattern. A single EEG epoch may contain useful information, but it often does not provide enough context to describe how the signal is

evolving. For this reason, short-term temporal context is commonly incorporated into seizure detection systems.

In deep sequential models, this context may be learned implicitly. In a feature-based framework, however, it can be introduced explicitly by stacking features from the current epoch together with features from several preceding epochs. This allows the classifier to base each decision not only on the current signal state, but also on its recent temporal development. The result is still a structured tabular input, but one that captures short-term EEG evolution more effectively than a single-window representation.

Taken together, the spectral, spatial, and temporal components form a representation that is richer than a simple per-window feature vector, while still remaining interpretable and computationally manageable.

3.3 From Window Classification to Event Detection

3.3.1 Class Imbalance

One of the main practical difficulties in seizure detection is class imbalance. In long EEG recordings, seizure epochs usually form only a very small proportion of the total data, while non-seizure background dominates the recording. If this imbalance is not addressed, a classifier may become biased toward the majority class and achieve apparently strong accuracy while failing to detect the minority class of real clinical interest.

A common response is to rebalance the training process by controlling the ratio of negative to positive samples, by weighting classes differently, or by combining both strategies. The key idea is not to remove background activity entirely, because the model still needs to learn the diversity of non-seizure EEG, but to prevent training from being driven almost entirely by the majority class. In the present project, this principle motivates a controlled negative-sampling strategy in the patient-specific training pipeline.

3.3.2 Thresholding and Post-Processing

Many classifiers do not output a hard seizure decision directly. Instead, they output a score or probability indicating how likely an epoch is to be seizure-related. A threshold is then applied to convert this continuous output into a binary decision. This threshold plays a central role because it determines the operating point of the detector. A lower threshold may improve sensitivity, but often at the cost of more false alarms. A higher threshold may reduce false alarms, but can increase missed detections or delay the alarm.

Raw thresholding alone is usually not enough for a clinically meaningful detector. EEG score sequences often contain isolated spikes caused by noise, ambiguity, or local fluctuations in the classifier output. If each spike is treated as a separate alarm, the false alarm burden can become unacceptable. For this reason, post-processing is commonly applied after thresholding.

Two ideas are especially important. First, smoothing can reduce short unstable fluctuations in the predicted seizure probability. Second, a minimum event-duration rule can suppress very brief positive runs that are unlikely to represent clinically meaningful seizure events.

Together, these steps help convert noisy window-level decisions into more stable event-level detections.

3.3.3 Event Formation

The distinction between window classification and event detection is central to this project. A detector is not clinically useful simply because it predicts many short windows correctly. It must identify seizure events as events. This means that predicted positive windows must be merged into alarm periods, and these alarm periods must then be compared with annotated seizure intervals.

Event formation therefore acts as a bridge between machine learning output and clinical interpretation. It is the stage at which a sequence of per-window scores becomes a detector that can be judged in terms of seizure capture, false alarm burden, and response delay.

3.4 Event-Level Performance Metrics

Because the final target is event detection rather than isolated window classification, the evaluation metrics must reflect clinical use more closely than raw accuracy alone [4]. Three measures are especially important in this context.

The first is event sensitivity, which measures the proportion of true seizure events that are successfully detected:

$$Sensitivity_{event} = \frac{N_{detected}}{N_{true}}$$

where $N_{detected}$ is the number of seizure events detected by the system and N_{true} is the total number of annotated seizure events.

The second is the false alarm rate per hour, which measures how often the detector raises a false seizure event relative to the total recording duration:

$$FAR/h = \frac{N_{false_alarm_events}}{total_recording_hours}$$

This metric is important because even a highly sensitive detector may be difficult to use in practice if it produces too many false alarms during long-term monitoring [4].

The third is detection delay, which measures how long the system takes to raise its first alarm after the clinical onset of a seizure:

$$d_i = t_{first_alarm,i} - t_{onset,i}$$

where d_i is computed only for detected seizures. Mean and median delay summarise the timing behaviour of the detector across seizure events.

These metrics capture three clinically important questions: Did the detector catch the seizure? How often did it alarm incorrectly? How quickly did it respond after onset? For this reason, they provide a more meaningful basis for evaluation than classification accuracy alone. The emphasis on event-level metrics in the present work follows directly from this clinical interpretation.

3.5 Classifier Families Considered

The project compares three classifier families that are well suited to structured EEG feature vectors: SVM with RBF kernel, Random Forest, and XGBoost. These models were selected not to represent every possible machine learning approach, but to provide a meaningful benchmark across classical and modern tabular classifiers. Their roles in the study are grounded in the original model-selection design of the project.

3.5.1 Support Vector Machine with RBF Kernel

Support Vector Machines remain important in seizure detection because they are closely associated with classical feature-based EEG classification [3], [8]. The main idea of an SVM is to find a decision boundary that maximises the margin between classes. When the data are not linearly separable, a kernel function can be used to project the data into a higher-dimensional space in which nonlinear separation becomes possible.

The radial basis function (RBF) kernel is widely used because it can model nonlinear decision boundaries effectively [8]. For EEG feature vectors, this is attractive because seizure and non-seizure patterns are rarely separated by a simple linear rule. SVMs are therefore a natural benchmark in feature-based seizure detection, especially when combined with appropriate standardisation and class-balancing strategies.

3.5.2 Random Forest

Random Forest is an ensemble learning method based on many decorrelated decision trees [7]. Each tree is trained on a variation of the training data, and the final decision is obtained by combining the outputs of the individual trees. This bagging-based strategy tends to improve robustness and reduce overfitting compared with a single decision tree.

Random Forest is particularly suitable for the present project for several reasons. It can model nonlinear interactions, handles heterogeneous feature sets well, and does not require aggressive feature scaling. It also supports feature-importance analysis, which is valuable in a patient-specific and clinically interpretable setting. Its main limitation is that large forests can lead to larger model size and a heavier deployment footprint, which becomes important when deployment-oriented analysis is considered.

3.5.3 XGBoost

XGBoost is a gradient-boosted tree method designed for efficient and strong performance on structured data [9]. Unlike Random Forest, which averages many independently grown trees, boosting builds trees sequentially so that each new tree focuses on correcting the errors made by the previous ensemble. This often produces a strong classifier, especially for tabular data with nonlinear interactions.

In the present context, XGBoost serves as a modern tabular baseline that complements the more classical SVM and the bagging-based Random Forest. It is relevant because seizure detection based on engineered EEG features is still a structured-data problem, and boosted trees are often among the strongest methods in such settings. At the same time, boosted

models may be more sensitive to hyperparameter choices and can be less straightforward to interpret at the global feature level than Random Forest.

Overall, these three classifier families provide a balanced comparison. SVM represents the classical feature-based tradition, Random Forest offers a strong interpretable ensemble baseline, and XGBoost adds a powerful modern boosting approach for structured EEG feature classification.

4. Dataset, Experimental Setup, and Validation Protocol

4.1 Dataset and Study Cohort

This project uses the CHB-MIT scalp EEG database, a widely used public benchmark for seizure detection research [1]. The database is particularly suitable for this study because it provides annotated scalp EEG recordings from patients with epileptic seizures and has been used extensively in both classical and recent machine learning studies.

The present work focuses on a 10-patient subset, namely chb01 to chb10, under a patient-specific evaluation setting. For each patient, seizure recordings are analysed together with associated non-seizure background recordings so that the detector is evaluated on both seizure capture and false alarm behaviour. Within this final study cohort, the selected system is evaluated over 580.57 h of EEG containing 55 annotated seizure events.

This subset is smaller than the full 24-patient setting reported by Shoeb and Guttag [1], and this should be stated clearly. However, it is still large enough to support a meaningful patient-specific benchmark, to expose both strong and weak patient cases, and to support structured analysis of sensitivity, false alarms, and detection delay.

4.2 Implementation Environment

The full pipeline is implemented in Python in a Jupyter-based research environment. The implementation is organised into modular stages covering data loading, preprocessing, feature extraction, validation, model selection, visualisation, and export. This modular structure supports both repeatable experimentation and later extension of the pipeline.

The main software stack includes MNE-Python for EEG file handling and signal processing, NumPy and SciPy for numerical and signal-processing operations, pandas for structured data handling, scikit-learn for SVM and Random Forest models, XGBoost for boosted tree classification, joblib for serialisation and caching, and standard plotting libraries for visualisation. These tools are not important only as software choices; together they provide a reproducible environment for evaluating the same feature pipeline and classifier configurations across multiple patients and validation folds.

4.3 Experimental Configuration

The experimental setup is defined by a fixed set of signal-processing, feature-representation, training, and decision parameters. Rather than treating these settings as unrelated constants, it is more useful to group them by their role in the pipeline.

First, the signal-processing stage defines the basic representation of the EEG. Recordings are segmented into fixed-length epochs, band-pass filtered, and transformed into a frequency-based feature representation. Second, the feature stage defines the structure of the input vector, including multichannel spectral features, synchrony-related spatial information, and short-term temporal context. Third, the training stage specifies how class imbalance is controlled within patient-specific model fitting. Finally, the decision stage defines how output probabilities are converted into event detections through threshold selection and post-processing.

In the final selected configuration, the system uses 2 s epochs, a 0.5-50 Hz band-pass filter, spectral energy features across 20 contiguous 2 Hz bins from 0 to 40 Hz, three synchrony features, and three previous epochs of temporal history. Decision thresholds are selected from a fixed grid of candidate values, while short positive runs are suppressed during post-processing to reduce unstable false alarms. The patient cohort for the final benchmark is restricted to chb01-chb10. These settings were not chosen arbitrarily; they were selected through the staged search and model-selection procedure reported later in Chapter 5.

4.4 Reproducibility and Feature Caching

Reproducibility is an important practical concern in a feature-based EEG pipeline because repeated experiments can easily become inconsistent when preprocessing settings, feature definitions, or temporal-history settings are changed over time. To reduce this risk, the project uses configuration-bound feature caching.

For each patient, extracted feature matrices are stored together with metadata linked to the active experimental configuration. This has two advantages. First, it avoids unnecessary repeated feature extraction from the original EDF recordings. Second, and more importantly, it reduces the chance of silently mixing feature matrices produced under incompatible settings. In other words, the cache is not only a speed optimisation; it also acts as a form of configuration control for the research workflow.

The stored metadata records the key properties needed to interpret the cache correctly, including feature dimensionality, temporal-history settings, epoch structure, and cache version information. This makes the pipeline easier to audit, rerun, and extend in later work.

4.5 Patient-Specific Validation Protocol

The central evaluation procedure is patient-specific leave-one-seizure-out (LOSO) validation. This protocol is designed to preserve patient specificity while still preventing the held-out seizure event from influencing model selection [1]. It therefore addresses one of the main methodological risks in seizure detection, namely optimistic evaluation caused by leakage between training and testing data.

For a patient with N seizure files, the validation is organised as an outer LOSO loop. In each outer fold, one seizure file is held out for testing. The non-seizure background files are shuffled and partitioned into N background subsets so that each outer fold also has an associated background test portion. The remaining seizure and background files then form the outer training pool.

Within this outer training pool, an additional validation split is created for threshold selection. One seizure file together with a subset of background data is reserved as inner validation data, while the remaining training data are used for feature selection and model fitting. Under this design, the held-out seizure file in the outer fold is never used during feature selection, threshold selection, or classifier fitting. This separation is important because it reduces the risk of indirect optimisation on the final test seizure event.

The result is a protocol that is stricter than a naive epoch-level split, while still being appropriate for patient-specific modelling. It allows the project to evaluate whether the detector generalises to an unseen seizure event from the same patient rather than merely to unseen windows from the same recording.

4.6 Fold-Wise Threshold Selection

In seizure detection, the decision threshold is part of the detector design, not merely a minor implementation detail. A classifier may produce useful probability scores, but its clinical behaviour depends strongly on where the threshold is placed. A threshold that is too low may increase false alarms substantially, while a threshold that is too high may miss seizures or delay their detection.

For this reason, the final system does not use one fixed threshold for all validation folds. Instead, a threshold is selected separately within the validation portion of each LOSO fold. Candidate thresholds are evaluated according to a ranked clinical criterion. Priority is first given to thresholds that achieve at least a minimum acceptable validation sensitivity. Among those candidates, lower false alarm rate is preferred, and then lower median detection delay is preferred. If no candidate threshold satisfies the minimum sensitivity condition, a fallback ranking is used.

This strategy reflects an important clinical trade-off. Thresholds are not chosen purely to minimise false alarms, because that could lead to missed seizures. Nor are they chosen purely to maximise sensitivity, because that could make the detector unusable in long recordings. Instead, the threshold policy is designed to preserve seizure capture while still controlling alarm burden and timing behaviour.

4.7 Validation-to-Deployment Workflow

The project distinguishes clearly between the role of validation models and the role of deployment-oriented patient models. During LOSO evaluation, the aim is to estimate how well the selected configuration generalises to unseen seizure events from the same patient. This stage provides evidence about model behaviour under retrospective testing conditions.

Once the final configuration has been selected, one deployment-oriented model is retrained for each patient using all available retrospective data for that patient. This means that the final patient model is not identical to any single validation-fold model. Instead, validation is used to justify the choice of configuration, while final retraining is used to produce the most informed patient-specific model that could later be exported, analysed, or carried forward to implementation studies.

This separation is conceptually important. It avoids confusion between model selection and final deployment training, and it makes the later export and profiling stages easier to interpret. The result is a workflow in which validation supports credibility, while retraining supports practical readiness for future deployment-oriented work.

5. Proposed Framework and Model Selection Strategy

5.1 Overall Framework

The proposed system is a patient-specific seizure detection framework designed to move from raw scalp EEG recordings to clinically interpretable event-level decisions. At a high level, the framework contains three connected stages: signal preparation and feature construction, fold-wise model selection under patient-specific validation, and post-processed event detection with deployment-oriented outputs.

In the first stage, EDF recordings and summary annotations are read, the target bipolar channels are aligned, and the multichannel EEG is filtered and segmented into fixed-length epochs. A structured feature vector is then constructed for each epoch by combining spectral, spatial, and short-term temporal information. In the second stage, candidate model configurations are compared under the patient-specific LOSO protocol described in Chapter 4, and the final configuration is selected through a staged search process. In the third stage, predicted probabilities are smoothed, thresholded, and converted into event detections, which are then evaluated using event sensitivity, false alarm rate per hour, and detection delay.

This modular structure is intentional. It allows the framework to be analysed at more than one level: as a signal-processing pipeline, as a model-selection workflow, and as a deployment-oriented detector that produces patient-level outputs rather than only validation scores.

5.2 Feature Instantiation for the Present Study

Based on the methodological foundations introduced in Chapter 3, the feature representation used in the present study is instantiated as a compact spectral-spatial-temporal vector for each patient-specific EEG epoch context.

The base feature representation contains 443 features. Of these, 440 are spectral features derived from 22 bipolar EEG channels with 20 spectral bins per channel, while the remaining 3 features capture inter-channel synchrony. This base representation is then extended with short-term temporal context by stacking the current epoch together with the previous three epochs. As a result, the final classifier input has a dimensionality of 1772.

This representation was selected because it provides a useful balance between richness and interpretability. Each element of the vector can be traced directly to a channel, a frequency interval, and a time position. This is important not only for model training, but also for later feature selection, feature-importance analysis, and patient-level export.

5.3 Hierarchical Configuration Search

The final framework was not produced from a single fixed configuration chosen in advance. Instead, model development followed a hierarchical search strategy in two stages. This was

done to separate different design questions and to avoid changing too many factors at the same time.

5.3.1 Phase A: Feature-Extraction Configuration

The first stage focused on the feature-extraction configuration. At this stage, the aim was to determine the most suitable signal-processing and temporal-context settings before performing more detailed model optimisation. To isolate the effect of these choices, top-k was held fixed while alternative feature-extraction configurations were compared.

The candidate settings differed in temporal-history length, filter implementation, and filter order. These alternatives were evaluated under the same patient-specific LOSO procedure and compared using event-level performance. The candidate configurations and their validation results may be summarised in Table 3.

Table 3. Phase A search results for feature settings

Rank	History Epochs	Filter Method	Butterworth Order	Mean Sensitivity	Median FAR/h	Mean Delay (s)	Patients
1	3	butter_sos	4	0.955	0.2617	7.63	10
2	4	butter_sos	6	0.955	0.3319	7.33	10
3	4	butter_sos	4	0.955	0.3319	7.37	10
4	4	fir_zero	4	0.950	0.3536	7.79	9

From this stage, one feature-extraction configuration was selected as the working basis for later optimisation. The chosen setting used three previous epochs of temporal history together with Butterworth SOS filtering of order 4. This phase established the feature-construction settings to be carried into the second stage of the search.

5.3.2 Phase B: Top-k and Random Forest Optimisation

Once the feature-extraction configuration had been fixed, the second stage focused on top-k feature reduction and Random Forest hyperparameters. This separation was important because it allowed the study to first identify a stable feature pipeline and then optimise the final classifier within that pipeline.

The Phase B search considered the number of selected features together with the key Random Forest settings, including the number of trees, maximum depth, minimum leaf size, and feature-subsampling policy. Candidate configurations were again compared under the same validation protocol, and the search results are reported in Table 4.

Table 4. Phase B search results for the Random Forest

Rank	Top-k	Trees	Max Depth	Min Leaf	Max Features	Mean Sensitivity	Median FAR/h	Mean Delay (s)
1	30	500	12	1	sqrt	0.980	0.2455	10.64
2	30	500	None	2	sqrt	0.970	0.3811	11.29
3	50	300	None	1	sqrt	0.955	0.2617	7.63
4	30	300	None	1	sqrt	0.955	0.2670	10.29
5	50	500	12	1	sqrt	0.955	0.2806	7.86
6	50	500	None	2	sqrt	0.930	0.3413	6.86

The selected final configuration used the top 30 features together with a Random Forest of 500 trees, maximum depth 12, minimum leaf size 1, and square-root feature sampling. In this way, feature reduction became part of the final model-design logic rather than a separate compression step added at the end.

5.3.3 Rationale for the Two-Stage Search

The purpose of this hierarchical search was methodological as well as practical. The first stage asked which feature pipeline should be used. The second stage asked how that feature pipeline should be paired with a compact and effective final classifier. By separating these two questions, the framework avoids treating all hyperparameters as one undifferentiated search space.

This design also supports clearer interpretation. If performance changes in Phase A, the reason is more likely to be linked to feature extraction and temporal context. If performance changes in Phase B, the reason is more likely to be linked to feature selection or classifier structure. The result is a model-selection pathway that is easier to justify and easier to analyse.

5.4 Final Model Selection and Patient-Level Deployment

A useful distinction in this project is the separation between benchmarking, final model selection, and deployment-oriented retraining.

First, benchmarking is used to compare the main classifier families under one common patient-specific protocol. In this role, SVM with RBF kernel, Random Forest, and XGBoost are treated as candidate model families operating on the same structured feature framework. Second, final model selection identifies the configuration that provides the strongest overall event-level trade-off within the chosen validation design. Third, deployment-oriented retraining uses the selected configuration to train one final patient-specific model from all available retrospective data for that patient.

This distinction matters because the model used to evaluate generalisation is not conceptually the same as the model that would later be exported or deployed. LOSO validation provides evidence about how the selected configuration behaves on unseen seizure events from the same patient. Final retraining then produces the most informed patient-specific model possible under the retrospective data available.

Under the final framework, each patient-specific deployment bundle stores the trained Random Forest model together with the selected feature indices, feature names, the patient-specific threshold, and supporting metadata describing the input structure and training composition. This makes the framework more than a validation exercise. It produces concrete artefacts that can be inspected, exported, and carried forward into later implementation work.

In addition to the main top-30, 500-tree path used for the primary benchmark and deployment profiling, a supplementary lightweight branch was examined to test model reduction more directly. In that branch, a standard Random Forest with 300 trees and full depth on the top-50 feature set was compared with a compact 120-tree variant with maximum depth 10 and minimum leaf size 2. Separately, a compact C export was demonstrated on the smallest-node single-patient top-30 RF example as an embedded representation proof-of-concept. These two elements are therefore treated at different evidence levels: the 120-tree branch is a validated RF compression branch, whereas the compact export is only a single-patient feasibility example.

5.5 Interpretability and Deployment Outputs

An important aim of the framework is to produce outputs that support interpretation as well as prediction. For this reason, the final system is designed to generate several forms of deployment-oriented and clinically useful output.

First, the selected feature subset can be ranked and named directly, which allows the final model to be interpreted in terms of channel, frequency range, and temporal lag. Second, the framework supports clinical single-case visualisation by combining seizure probability, alarm state, and filtered EEG traces around the seizure interval. Third, the exported metadata preserves the exact feature and threshold context needed to reconstruct the patient-level inference setting in later software or firmware studies.

These outputs are not treated as optional extras. They are part of the research contribution of the framework itself. The project is therefore positioned not simply as the training of a patient-specific classifier, but as the construction of a more complete seizure detection workflow that connects model selection, interpretability, and deployment-oriented analysis in one coherent system.

6. Experimental Results and Discussion

6.1 Comparative Benchmark Across Candidate Models

The first result question is which classifier family provides the most suitable overall event-level trade-off within the proposed patient-specific framework. To answer this, SVM with RBF kernel, Random Forest, and XGBoost were compared under the same feature configuration and the same LOSO evaluation protocol.

The benchmark results show that Random Forest provides the most balanced overall performance among the three candidate models. SVM-RBF and Random Forest reach the same macro sensitivity of 0.955, but Random Forest reduces the median false alarm rate substantially, from 1.1788/h to 0.2767/h. XGBoost achieves the shortest mean delay at 4.53 s,

but its macro sensitivity falls to 0.930, and its false alarm rate remains higher than that of Random Forest. These results indicate that, within the present evaluation setting, Random Forest offers the strongest overall balance between seizure capture, alarm burden, and response time.

Table 5. Macro benchmark comparison of SVM-RBF, Random Forest, and XGBoost

Model	Macro Sensitivity	Median FAR/h	Mean Delay (s)	Median Delay (s)	Median Threshold
SVM-RBF	0.955	1.1788	8.35	4.5	0.7415
Random Forest	0.955	0.2767	7.36	7.5	0.4730
XGBoost	0.930	0.3419	4.53	3.5	0.7680

This comparison is important because it shows that final model choice should not be based on one metric alone. A shorter delay is attractive, but not if it is achieved at the cost of lower sensitivity. Likewise, matching sensitivity is not sufficient if the alarm burden remains much higher. For the present framework, Random Forest is therefore selected as the main classifier family for the final system because it provides the most favourable event-level trade-off rather than the best value on any single metric.

6.2 Final Model Validation and Feature Reduction

After selecting Random Forest as the preferred model family, the next question is whether explicit feature reduction improves or weakens overall detector behaviour. This was examined by comparing the full-feature Random Forest with the final top-30 Random Forest using pooled event statistics across the full study cohort.

The comparison shows that feature reduction does not merely compress the model; it also changes its detection behaviour. The full-feature model detects 52 of 55 seizures, while the top-30 model detects 53 of 55. At the same time, the false alarm rate decreases slightly from 0.3152/h to 0.3066/h. The main cost is a rise in mean detection delay from 6.69 s to 9.09 s. Within the present evaluation setting, this represents a reasonable trade-off because the reduced model improves event sensitivity slightly, lowers the false alarm burden slightly, and greatly reduces input dimensionality from 1772 to 30 features.

Table 6. Event-level comparison of the full-feature RF and the top-30 RF

Variant	TP Events	FN Events	FP Events	Event Sensitivity	FAR/h	Mean Delay (s)	Hours	Patients
RF full feature	52	3	183	0.9455	0.3152	6.69	580.57	10
RF top-30	53	2	178	0.9636	0.3066	9.09	580.57	10

This result is central to the project. It shows that top-k selection should not be viewed only as a deployment-driven compression step. In this study, it becomes part of the final model-design logic. The selected top-30 model is smaller, easier to interpret, and slightly stronger in pooled event terms than the full-feature alternative.

The final system was then evaluated under patient-specific LOSO validation across the full 10-patient study cohort. Across 580.57 h of scalp EEG containing 55 annotated seizures, the

selected model achieved a macro sensitivity of 0.98, a median false alarm rate of 0.2455/h, and a mean detection delay of 10.64 s. Expressed in a day-normalised form, this corresponds to approximately 5.89 false alarms per 24 h.

Table 7. Final 10-patient LOSO validation results for the selected model

Patient	Hours	True Seizures	Sensitivity	FAR/h	Mean Delay (s)	Median Delay (s)	Median Threshold
chb01	40.55	7	1.00	0.6658	4.29	2.0	0.499
chb02	35.27	3	1.00	0.2552	3.33	4.0	0.308
chb03	38.00	7	1.00	0.5263	4.00	2.0	0.672
chb04	156.06	4	1.00	0.4229	41.00	25.0	0.742
chb05	39.00	5	1.00	0.1795	12.40	6.0	0.482
chb06	66.73	10	0.80	0.0899	4.25	4.0	0.308
chb07	67.05	3	1.00	0.0895	15.33	18.0	0.690
chb08	20.01	5	1.00	0.7498	11.20	12.0	0.464
chb09	67.87	4	1.00	0.2357	7.50	6.0	0.690
chb10	50.02	7	1.00	0.1199	3.14	4.0	0.811
MACRO	580.57	55	0.98	0.2455	10.64	5.0	0.5855

The patient-level results show that performance is good overall but not uniform across all subjects. Nine of the ten patients achieve perfect seizure sensitivity, while chb06 reaches a sensitivity of 0.80 and is the only patient in the final cohort with missed seizures. In addition, the main weaknesses are distributed differently across patients. For chb04, the detector achieves full seizure capture but with a large mean delay of 41 s, indicating that the main issue is timing rather than missed events. For chb08, sensitivity also remains perfect, but the false alarm burden rises to 0.7498/h, indicating that non-ictal background activity in this patient is more easily confused with seizure structure.

Two complementary views of performance are used in this chapter and should be interpreted carefully. The macro summary averages performance across patients and uses the median false alarm rate across subjects, which prevents long recordings from dominating the summary. The pooled event comparison instead aggregates all seizure events and false alarm events over all hours of data. These two views do not answer exactly the same question, and both are useful in understanding the final detector.

6.3 Interpretation and Clinical Case Illustration

An important goal of the framework is to remain interpretable after model selection. To examine this, the top-20 feature importance ranking was extracted for a representative patient-specific deployment model.

The most influential features are concentrated mainly in the theta and delta ranges, especially around 4-6 Hz and 2-4 Hz, and are associated with channels such as F4-C4, F3-C3, FT9-FT10, C4-P4, and FZ-CZ. Important features appear not only at the current time step but also at earlier time lags, which supports the view that short-term temporal evolution contributes meaningfully to seizure detection in the final system. These patterns are consistent with the

broader design logic of the framework, in which spectral, spatial, and temporal information are all expected to contribute to discrimination.

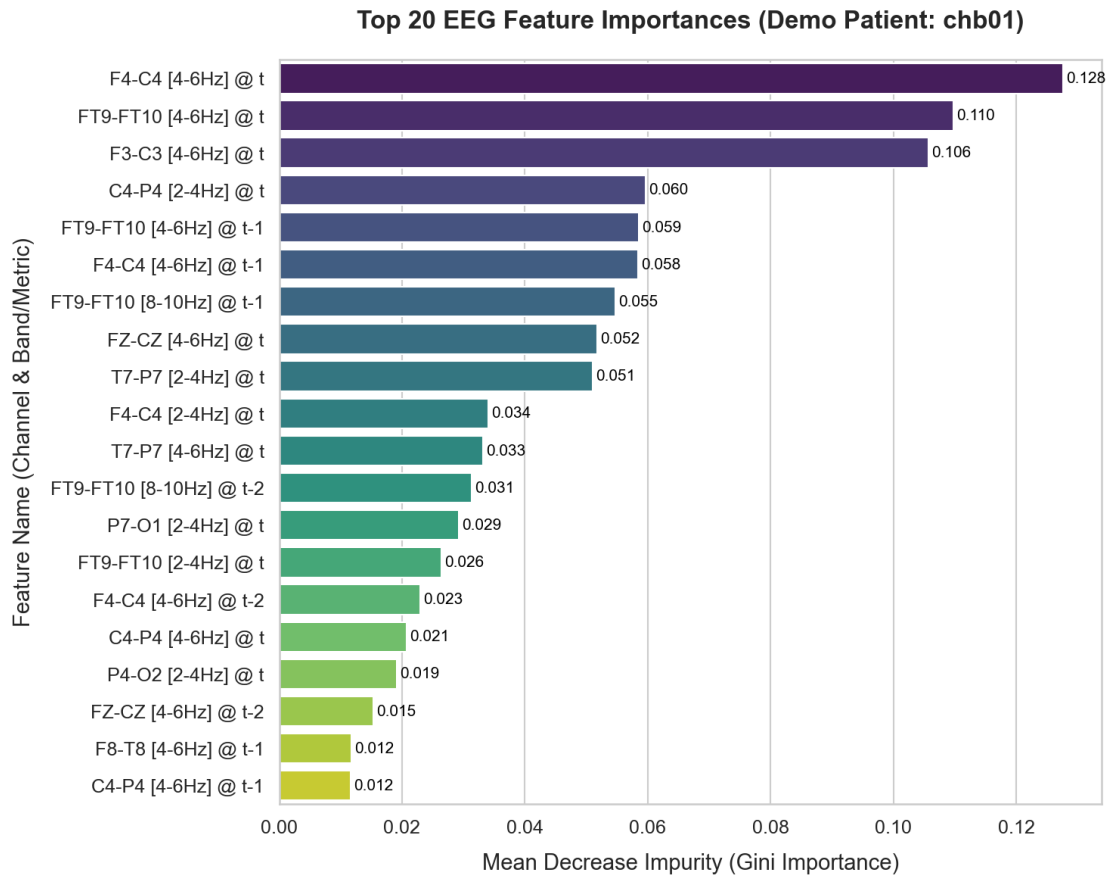


Fig. 3. Top-20 feature importance ranking for a representative patient-specific Random Forest model.

These results should be interpreted carefully. They do not establish one universal neurophysiological rule for all patients. Rather, they show that the selected feature-based model can be interpreted in clinically plausible terms at the level of a concrete patient-specific detector. This remains one of the main advantages of the final Random Forest approach over a less transparent end-to-end alternative.

A second interpretability result comes from the single-case clinical visualisation generated for chb01_03.edf. In this representative example, the detector raises an alarm at seizure onset, with no false alarm before the event. The plot combines filtered EEG traces, predicted seizure probability, and the final alarm state in one view.

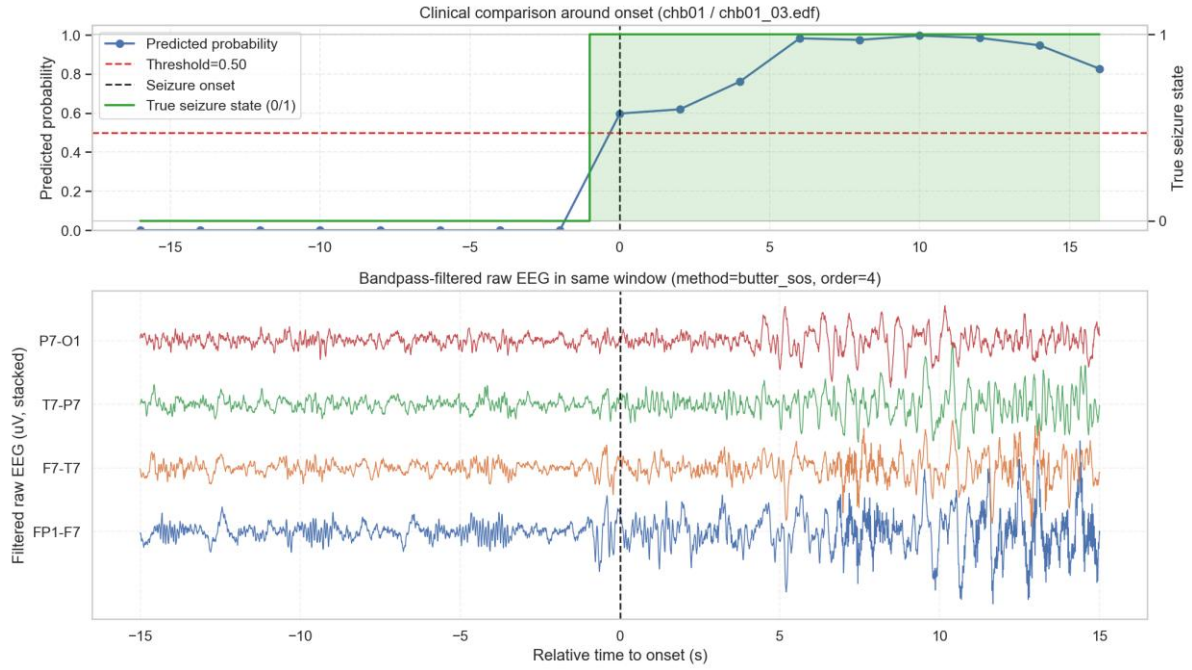


Fig. 4. Representative single-case visualisation showing filtered EEG, seizure probability, and alarm state around seizure onset.

This example should be read as an illustrative case rather than as a summary of average performance. Its value is that it links the final detector output back to a clinically readable signal context. In this sense, the framework supports not only benchmarking but also case-based inspection, debugging, and future clinician-facing explanation.

6.4 Error Analysis and Refinement Branch

Although the primary benchmark results are strong overall, they also reveal two clear patient-level weaknesses: delayed detections in chb04 and an elevated false alarm burden in chb08. To investigate these cases in more detail, a separate clinical refinement branch was developed after the main benchmark had been established.

This refinement branch uses its own reconciled baseline and should not be mixed directly with the primary headline benchmark. Within the branch, the reference baseline is macro sensitivity 0.9778, macro FAR/h 0.3907, and mean delay 9.96 s across 54 seizures and 564.56 h of data. These values differ from the main benchmark because they come from a different evaluation snapshot. The branch results should therefore be interpreted relative to this internal baseline only.

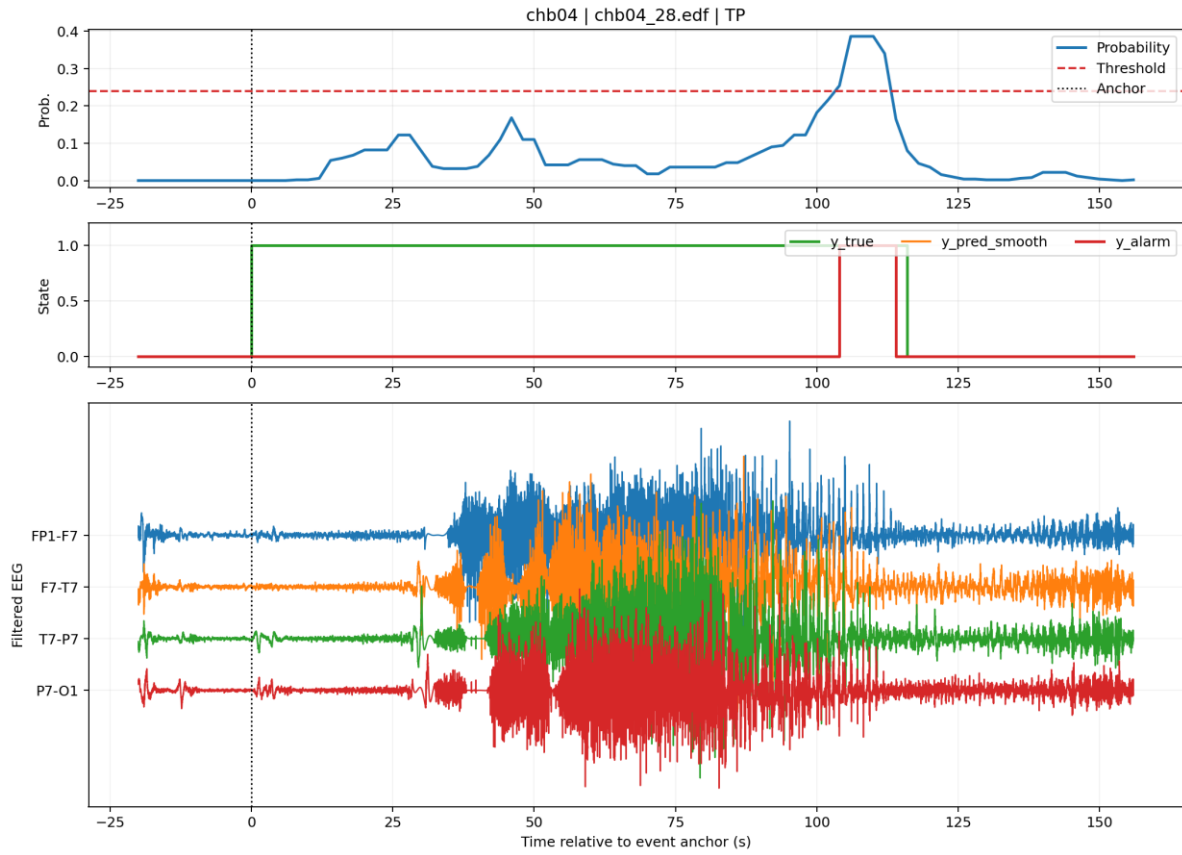


Fig. 5. Delayed true-positive example from chb04 in the clinical refinement branch.

Figure 5 shows the main chb04 failure mode clearly. The seizure is captured at event level, but the alarm rises only after seizure onset, so the weakness is primarily detection timing rather than seizure capture.

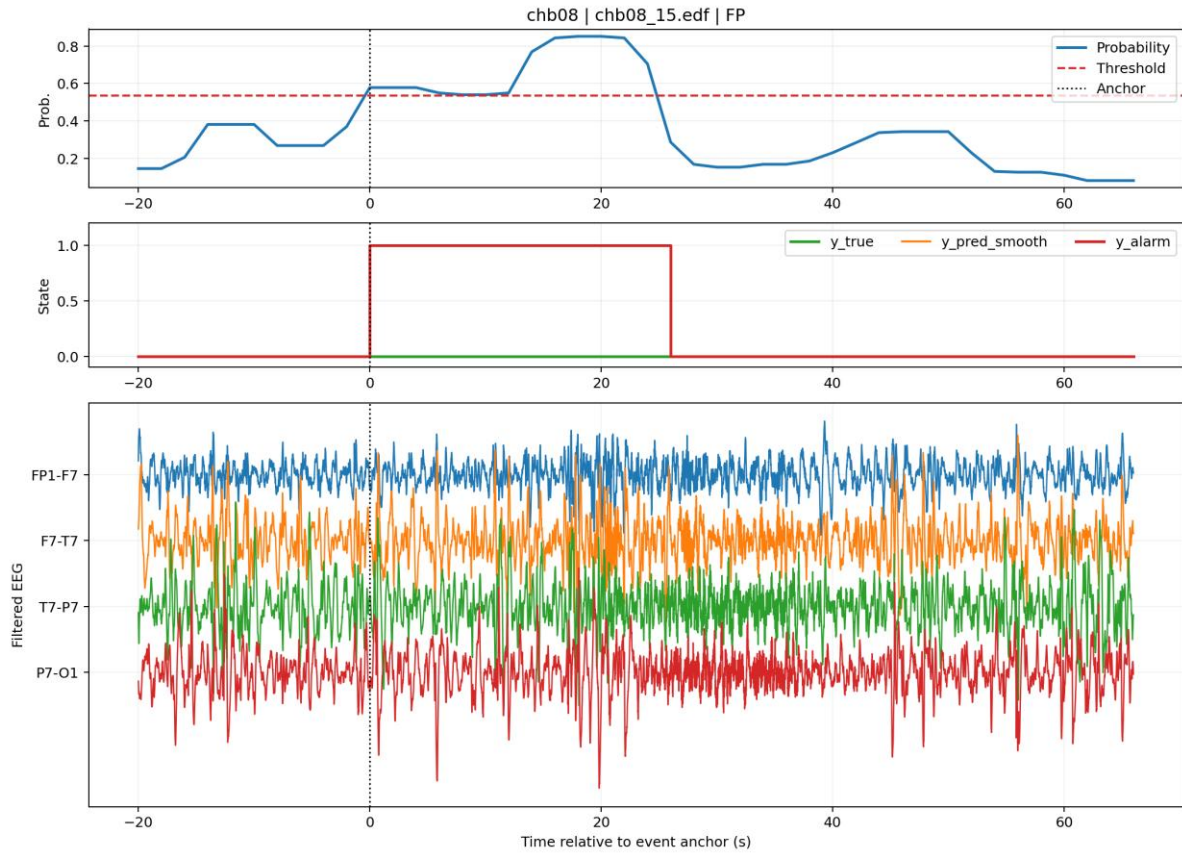


Fig. 6. False-positive example from chb08 in the clinical refinement branch.

Figure 6 shows the complementary chb08 problem. The detector crosses threshold on a non-ictal segment, which explains why this patient remains the clearest false-alarm burden case in the refinement branch.

The refinement process then developed through five iterations. V1 was mainly descriptive and focused on structured review of delayed detections and false positives. V2 introduced proxy augmentation and produced the strongest accepted result within the branch: macro sensitivity remained at 0.9778, while FAR/h decreased to 0.2575 and mean delay improved slightly to 9.42 s. V3 tested a smaller proxy subset and reduced FAR/h slightly further to 0.2551, but macro sensitivity fell to 0.9528 because one seizure in chb04 was missed. V4 added back a small amount of amplitude information and recovered part of the lost sensitivity, but delay worsened and the overall trade-off did not improve sufficiently. V5 adjusted threshold policy only and showed that more aggressive timing-oriented thresholding could maintain sensitivity, but at the cost of a severe increase in false alarms, with FAR/h rising to 1.2423.

Table 8. Summary of the clinical error analysis and refinement path from V1 to V5

Iteration	Main Change	Representative Result	Macro Sens.	FAR/h	Outcome
V1	Structured review of chb04 delay and chb08 FP cases	Failure modes identified; FP labels mostly uncertain	-	-	Diagnostic starting point
V2	Split proxy augmentation from suppression	proxy_augmented_rf	0.9778	0.2575	Best accepted branch result
V3	Proxy subset ablation	proxy_hybrid_rf	0.9528	0.2551	Rejected: one chb04 seizure missed
V4	Minimal add-back to recover chb04 sensitivity	proxy_hybrid_plus_count_rf	0.9667	0.2810	Rejected: chb04 delay worsened
V5	Timing-only threshold refinement	delay-first branch	0.9778	1.2423	Rejected: severe FP rise

These branch results support a focused practical conclusion. Proxy augmentation appears to be the most promising refinement direction among the variants tested, while global threshold lowering is not acceptable because it increases false alarms too sharply. More broadly, the refinement branch shows that local improvements on difficult patients are not sufficient on their own. A refinement must remain acceptable at cohort level before it can be treated as a meaningful improvement to the detector.

6.5 Deployment-Oriented Profiling

The final stage of analysis examines the detector from a deployment-oriented perspective. This includes inference latency, model size, export behaviour, and the structure of the patient-specific deployment models.

For a representative deployment model, the runtime profile shows a mean latency of 80.99 ms, a median latency of 70.89 ms, a P95 latency of 131.05 ms, and a throughput of 12.3 frames per second under the Python notebook environment. The same model uses 30 selected input features, 500 trees, and 36,706 total nodes, with a serialised footprint of 2.99 MB.

Table 9. Runtime and model-size profile for a representative deployment model

Metric	Value	Notes
Test epochs	1000	repeated single-epoch inference benchmark
Mean latency	80.99 ms	Python environment benchmark
Median latency	70.89 ms	central tendency
P95 latency	131.05 ms	high-percentile latency
Throughput	12.3 FPS	epochs per second
Input dimension	30	after top-k feature selection
Number of trees	500	final Random Forest
Total nodes	36,706	demo patient model
Decision nodes	18,103	demo patient model
Leaf nodes	18,603	demo patient model
Max tree depth	12	bounded tree depth
Serialized model footprint	2.99 MB	joblib dump size for demo patient model

These results must be interpreted with care. The latency measurements were obtained in a Python software environment and are therefore not direct embedded-device measurements. They nevertheless provide a useful baseline for estimating the computational scale of the final detector after feature reduction.

The model export analysis leads to a more critical engineering conclusion. Direct export of the 500-tree Random Forest to an inline C header is successful, but the generated header reaches approximately 3396.45 KB. This means that the project has clearly moved toward deployment-oriented implementation, but the present forest is still too large to be treated as a finished microcontroller-ready model for a strict low-memory target. In practical terms, the framework is deployment-oriented rather than fully embedded-ready.

The patient-specific deployment summaries reinforce this conclusion. While all exported models use the same reduced 30-feature input space, the tree structures remain non-trivial, and node counts vary substantially across patients. This means that feature reduction alone is not enough to guarantee lightweight embedded deployment. Further work will still be needed in pruning, compression, or alternative model representation before strict MCU implementation can be considered realistic.

Table 10. Summary of patient-specific deployment models and key structural parameters

Patient ID	Input Dim	Decision Threshold	Total Train Rows	Positive Ratio	RF Trees	Total Nodes
chb01	30	0.499	72,993	0.31%	500	36,706
chb02	30	0.308	63,479	0.14%	500	18,664
chb03	30	0.672	68,403	0.30%	500	30,636
chb04	30	0.742	280,914	0.07%	500	51,378
chb05	30	0.482	70,205	0.40%	500	39,088
chb06	30	0.308	120,122	0.07%	500	56,998
chb07	30	0.690	120,693	0.14%	500	40,504
chb08	30	0.464	36,011	1.28%	500	139,280
chb09	30	0.690	122,165	0.12%	500	24,408
chb10	30	0.811	90,041	0.25%	500	29,960

6.6 Lightweight Model Variant and Compact Embedded Representation

A supplementary lightweight branch was used to examine how far Random Forest compression could proceed before the performance penalty became too large. This branch should be interpreted as supplementary to the main benchmark because it used a different Random Forest configuration and a top-50 feature setting rather than the selected top-30, 500-tree model used for the main headline results.

Within that branch, the validated 120-tree compact Random Forest reduced the mean serialized size across the exported patient models in the lightweight branch from 3642.3 KB to 719.2 KB, which corresponds to a reduction of about 80.3%. The cost was a drop in macro sensitivity from 0.955 to 0.930 and an increase in FAR/h from 0.2617 to 0.4234, while mean delay improved slightly from 7.63 s to 6.63 s. This means that substantial RF compression is possible, but the trade-off is not negligible and the compact branch should not replace the main validated detector in the headline findings.

Table 11. Validated lightweight RF branch summary

Model	Trees	Input Features	Sensitivity	FAR/h	Mean Delay (s)	Serialized Size
Reference RF (300 trees, top-50)	300	50	0.955	0.2617	7.63	3642.3 KB (mean)
Compact RF (120 trees)	120	50	0.930	0.4234	6.63	719.2 KB (mean)

Note: Serialized sizes are means across the exported patient models in the lightweight branch.

A separate compact C export was also generated from the smallest-node single-patient top-30 RF example. In that case, a 100-tree model with 30 input features and 3454 total nodes was compressed into a compact C representation with a payload of 21164 bytes (20.67 KB). The purpose of this step was only to show that more aggressive embedded compression is technically feasible. It was not re-validated at cohort level and should not be interpreted as a replacement for the validated 120-tree branch.

Taken together, these results support a balanced engineering conclusion. Moderate Random Forest compression is feasible, but not without a measurable performance trade-off. A separate single-patient compact export also suggests that more aggressive embedded compression is technically possible, though only at proof-of-concept level. The present evidence therefore supports deployment feasibility, but it does not justify the claim that a fully embedded-ready final detector has already been achieved.

6.7 Chapter Summary

The results of this chapter support seven main conclusions.

First, patient-specific seizure detection remains a strong strategy for this problem under the present study design. Second, Random Forest provides the most balanced overall event-level trade-off among the three candidate model families. Third, top-30 feature selection improves the pooled event-level outcome slightly while also improving interpretability and reducing input dimensionality substantially. Fourth, sensitivity alone is not sufficient to judge detector usefulness; the cases of chb04 and chb08 show that delay and false alarm burden must be examined separately. Fifth, the refinement branch suggests that proxy augmentation is a promising direction for follow-on work, whereas more aggressive threshold-based changes do not generalise well. Sixth, the lightweight branch shows that average serialized Random Forest size can be reduced by about 80% with a 120-tree variant, but only with a moderate loss in sensitivity and a higher false alarm burden. Seventh, a separate single-patient compact C export suggests that further embedded compression is feasible at prototype level, but it should not be interpreted as a cohort-level validated detector. Taken together, the framework now supports benchmark-level evaluation, interpretation, error analysis, lightweight exploration, and deployment-oriented profiling, but the current exported models should still be described as deployment-oriented rather than fully embedded-ready.

7. Summary, Contribution, and Critical Reflection

7.1 Overall Summary

This project set out to develop a patient-specific scalp EEG seizure detection framework that was technically rigorous, clinically interpretable, and realistic about deployment constraints. Within the scope of the present study, that goal has largely been achieved. The final system implements a complete workflow that spans signal preparation, spectral-spatial-temporal feature construction, patient-specific LOSO validation, model selection, feature reduction, patient-level retraining, interpretability analysis, clinical visualisation, and deployment-oriented profiling.

In the main benchmark, the selected top-30 Random Forest achieved a macro sensitivity of 0.98, a median false alarm rate of 0.2455 per hour, and a mean detection delay of 10.64 s across 10 patients, 580.57 h of EEG, and 55 annotated seizures. In pooled event terms, the system detected 53 of 55 seizures. These results show that the final framework performs well overall while still leaving some clinically important weaknesses to be addressed, especially in delay-sensitive and false-alarm-sensitive cases.

A separate clinical refinement branch then extended the main study by examining difficult patient cases more closely. That branch is valuable because it turned local failure patterns into a more structured research question, but its findings should be interpreted as follow-on analysis rather than merged directly with the primary benchmark.

7.2 Main Contribution

The contribution of this project is best understood not as the invention of a new classifier, but as the construction of a stronger and more complete research workflow around patient-specific seizure detection. Relative to a simple baseline study, the present work makes six main contributions.

First, it develops a richer but still interpretable feature representation by combining multichannel spectral structure, spatial synchrony information, and short-term temporal context within one patient-specific framework. Second, it evaluates candidate model families under a common event-level LOSO protocol rather than relying on weaker or less comparable validation settings. Third, it shows that explicit top-k feature reduction can support both interpretability and performance, rather than serving only as a compression step. Fourth, it carries the final system beyond model selection into patient-level export, interpretability analysis, runtime profiling, and structured error analysis. Fifth, it adds a supplementary lightweight branch that provides validated Random Forest compression analysis at cohort level. Sixth, it includes a separate compact export feasibility example to illustrate the technical possibility of more aggressive embedded compression.

Taken together, these contributions place the project in an important middle ground. The work does not stop at a basic feature-based detector, but neither does it jump directly to a complex deep-learning claim without first establishing a strong benchmark. Instead, it builds an interpretable and extensible system that can support more rigorous follow-on research.

7.3 Critical Reflection and Limitations

The main strengths of the project lie in its modular design, its stronger-than-naive validation strategy, its focus on clinically meaningful event-level metrics, and its effort to connect model performance with interpretability and deployment-oriented analysis. These features make the work more credible than a simple notebook demonstration and provide a clearer basis for later extension.

At the same time, several limitations remain clear. The final benchmark covers only 10 patients rather than the full CHB-MIT cohort. The evaluation is retrospective and offline rather than prospective. The framework is patient-specific, so it does not address the cold-start problem for unseen patients. Artifact robustness is only partly handled through filtering and post-processing, and no dedicated artifact classifier is included. In addition, although model export was demonstrated successfully, the current direct C export remains too large for strict embedded targets, which means that deployment readiness is still partial rather than complete.

A further limitation concerns interpretation of the later refinement work. The clinical refinement branch used a separate reconciled baseline and should therefore be understood

within that branch only. More broadly, repeated focus on difficult cases such as chb04 and chb08 may introduce selection bias even when direct train-test leakage is controlled. These limitations do not cancel the value of the present study, but they do define the correct boundary of its claims.

7.4 Final Conclusion

The main conclusion of this report is that a patient-specific, feature-based, event-level seizure detector remains a promising research direction for scalp EEG. Within the scope of the present benchmark, the top-30 Random Forest provides the best overall balance among the tested models. It combines strong seizure capture with a relatively low false alarm burden, preserves interpretability, and supports a more deployment-oriented pipeline than a full high-dimensional feature model.

A second conclusion is that useful progress in seizure detection depends on more than classifier choice alone. It also depends on validation discipline, threshold design, event formation, false alarm control, patient-level model handling, and honest engineering assessment. The present work contributes most strongly by bringing these elements together in one coherent framework. The supplementary lightweight branch reinforces this point: meaningful compression is feasible, but not without performance trade-offs.

A further practical result is that a separate single-patient compact C export reduces the embedded payload to 20.67 KB. This should be read as a deployment-oriented representation prototype rather than as a one-to-one replacement for the validated 120-tree RF branch, and the lightweight branch should not be interpreted as rewriting the main benchmark.

For that reason, the project should be seen as a solid research prototype rather than a finished clinical or embedded product. It provides a stronger benchmark, a clearer understanding of remaining failure modes, a credible lightweight exploration path, and a more realistic starting point for the follow-on thesis stage.

8. Follow-On Thesis Development Plan

8.1 Thesis-Level Aim

The next stage of this work is not simply to continue running more experiments. Its aim is to develop the current research prototype into a more fully validated, more deployment-aware, and more critically benchmarked thesis system. In practical terms, this means strengthening the evidence base of the current patient-specific detector, improving its engineering realism for embedded use, and extending the study in a way that responds directly to the remaining limitations identified in Chapter 7.

The thesis plan therefore has three primary priorities. The first is broader validation on a larger patient cohort. The second is embedded-oriented model compression and hardware-grounded implementation analysis. The third is focused refinement of the remaining hard-case failure modes, especially the trade-off between delayed detections and false alarms. A further comparison with selected deep sequential baselines is retained as a valuable secondary

extension, but it should be treated as conditional on the stability of the earlier benchmark and implementation work.

8.2 Primary Priority 1: Broader Validation on a Larger Cohort

The first priority of the follow-on thesis should be to strengthen the validation evidence of the present detector. The current benchmark on 10 patients is meaningful, but it is still narrower than the full CHB-MIT cohort and therefore limits the strength of the project's broader claims. The most direct next step is to extend the evaluation to more patients while preserving the same patient-specific protocol, event-level metrics, and final feature framework.

This work package should focus on four core tasks. First, the benchmark should be extended beyond chb01-chb10. Second, the feature caches should be rebuilt under the selected final configuration so that the enlarged benchmark remains internally consistent. Third, the LOSO evaluation should be rerun using the same event-level criteria already established in the present report. Fourth, the stability of the main conclusions should be examined on the larger cohort, especially the choice of Random Forest and the use of top-30 feature reduction.

A second public dataset may be considered later if time and compatibility allow, but this should be treated as a secondary extension rather than a core deliverable. The primary objective is first to establish whether the present findings remain credible on a broader CHB-MIT cohort. The expected outcome of this work package is a stronger and more statistically credible validation chapter for the thesis.

8.3 Primary Priority 2: Embedded-Oriented Compression and Hardware Evaluation

The second priority is to move the current detector closer to realistic embedded implementation. The present project has already shown that patient-level export is feasible, but it has also shown clearly that the current 500-tree Random Forest remains too large for a strict low-memory embedded target in its present form. The next thesis stage should therefore focus on model compression and hardware-grounded evaluation rather than on software-only export claims.

This work package should begin with controlled model simplification. Smaller forests, shallower trees, or more compact model representations should be tested while monitoring the resulting event-level loss in sensitivity, false alarm rate, and delay. The key question is not whether the model can be made smaller in principle, but whether it can be made smaller without undermining the detector's clinical usefulness.

Once a compressed model is identified, the next step should be evaluation on one concrete embedded target, such as an STM32-class board. The essential measurements are RAM use, flash footprint, and runtime latency under a realistic implementation path. Porting the complete feature extractor to C may be pursued gradually, beginning with the most important components of the final top-30 pipeline, but full feature-pipeline porting should be treated as a later-stage objective rather than a required first milestone. The expected outcome of this work package is a deployment chapter grounded in measured hardware behaviour rather than software-only estimates.

8.4 Primary Priority 3: Focused Refinement of Hard-Case Failure Modes

The third priority is to improve the detector on the most informative difficult cases without weakening performance at cohort level. The current report has already identified a specific tension: some patients are mainly limited by delayed detection, while others are mainly limited by elevated false alarms. In the present cohort, chb04 and chb08 provide the clearest examples of this trade-off.

The next stage of refinement should therefore be more focused than the earlier exploratory branch. Rather than reopening a broad search over many unrelated heuristics, the thesis should build mainly on the most promising refinement direction already identified, namely proxy-feature augmentation. Additional changes should be targeted and hypothesis-driven, with particular emphasis on improving onset timing without globally lowering thresholds in a way that would reintroduce a heavy false alarm burden.

This work package should also strengthen the structure of the error analysis itself. False-positive review categories should be made more informative where possible, and any new refinement should continue to be tested against both the difficult focus cases and the wider cohort-level guard conditions. The expected outcome is a more rigorous failure-analysis chapter that not only describes errors, but also tests controlled routes for improving the delay versus false-alarm trade-off.

8.5 Secondary Extension: Comparison with Selected Deep Sequential Baselines

A comparison with modern deep sequential models remains an important thesis extension, but it should be treated as secondary to the three priorities above. The main value of such a comparison is not to assume that deep learning must be better, but to test whether a more complex sequential model offers a meaningful improvement under the same patient-specific protocol and event-level evaluation framework.

To keep this extension realistic, the comparison should be limited to one or two carefully chosen baselines rather than several unrelated deep architectures. A suitable plan would be to compare the final feature-based Random Forest against one CNN-based sequential baseline and one lightweight temporal model, such as a compact CNN-BiLSTM or lightweight temporal CNN. These models should reuse the same patient partitions wherever possible so that the comparison remains fair.

The comparison should examine not only event sensitivity, false alarm rate, and delay, but also training cost, interpretability, and deployment cost. Its purpose is to answer a focused research question: does a more complex sequential model provide enough gain to justify its additional complexity when compared against the interpretable top-30 Random Forest benchmark already established in the present work.

8.6 Proposed Thesis Timeline

The follow-on timeline should reflect the dependencies between these tasks. The first stage, covering May and early June, should focus on cohort expansion and rerunning the broader

patient-specific benchmark. This is the natural starting point because later comparison and compression work depend on a stable and credible benchmark configuration.

The second stage, covering June and July, should focus on compression studies and hardware-grounded implementation analysis. Once the final benchmark setting has been confirmed on a broader cohort, model simplification and embedded testing can proceed on a more stable basis.

The third stage, mainly in July, should focus on hard-case refinement and error analysis. By this point, the benchmark and deployment assumptions should be clearer, which will make refinement experiments easier to interpret. A deep-model comparison may then be added in late July or August if the earlier stages have stabilised and sufficient time remains.

The final stage, in August, should consolidate the benchmark, compression, refinement, and any selected comparison results into the final thesis chapters, figures, tables, and appendices. This sequence is more realistic than treating all work packages as equal and simultaneous tasks from the beginning.

8.7 Expected Thesis Outcome

If completed successfully, the follow-on thesis should deliver four main outcomes. First, it should provide a broader and more credible patient-specific benchmark than the current 10-patient study. Second, it should produce a more realistic deployment-oriented implementation through measured compression and hardware evaluation. Third, it should clarify the remaining trade-off between delayed detection and false alarm control on the hardest patient cases. Fourth, it should establish whether the current interpretable feature-based detector remains competitive when compared fairly against selected deep sequential baselines.

In this way, the thesis would respond directly to the main limitations of the current report: restricted cohort size, only partial deployment readiness, unresolved hard-case behaviour, and the absence of a like-for-like comparison with stronger sequential baselines. The intended outcome is therefore not just a larger version of the present project, but a more complete and more defensible thesis contribution.

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Appendix A. Safety Statement and Risk Analysis

This project uses publicly available, anonymised scalp EEG data and performs offline computational analysis only. No human participants are recruited directly, and no clinical intervention is performed. Therefore, the main risks are not laboratory-electrical risks but research-practice risks:

1. accidental disclosure of derived files or data paths through insecure storage;
2. loss of data or results due to poor version control or notebook overwriting;
3. incorrect clinical interpretation if algorithmic outputs are over-claimed beyond the available evidence;
4. over-reliance on retrospective validation without prospective deployment testing.

The project mitigates these risks by:

1. using only de-identified publicly available EEG data;
2. storing outputs locally and separating cached data from final exported artifacts;
3. reporting the system as a research prototype rather than a clinical device;
4. explicitly distinguishing between validation evidence and deployment readiness.

No custom hardware was built during the EE6019 stage of the project, so electrical safety risks were minimal in the present semester. However, if embedded deployment is pursued during the summer, standard laboratory and embedded-development safety procedures will need to be followed.

Appendix B. Reproducibility Notes and Key Parameters

B.1 Main Software Stack

1. Python / Jupyter Notebook
2. MNE-Python
3. NumPy
4. SciPy
5. pandas
6. scikit-learn
7. XGBoost
8. matplotlib
9. seaborn
10. joblib

B.2 Core Final Parameters

1. patient subset: chb01-chb10
2. epoch length: 2 s
3. band-pass filter: 0.5-50 Hz
4. spectral bins: 20 bins from 0 to 40 Hz
5. synchrony pairs: 3
6. history epochs: 3
7. base feature dimension: 443
8. stacked feature dimension: 1772
9. final top-k: 30
10. final classifier: Random Forest
11. number of trees: 500
12. maximum depth: 12
13. minimum samples per leaf: 1
14. max features: sqrt
15. probability smoothing: median filter, kernel size 5
16. minimum alarm duration: 3 epochs

B.3 Original Final Macro Validation Result

These values are the headline results from the original final benchmark notebook and remain the main benchmark reference used in Chapters 5-7.

1. sensitivity: 0.98
2. median FAR/h: 0.2455
3. mean delay: 10.64 s
4. median delay: 5.0 s
5. total hours: 580.57
6. total seizures: 55

B.4 Original Final Exported Patient-Level Information

For each patient deployment bundle, the final notebook records:

1. patient ID;
2. trained Random Forest estimator;
3. top-30 feature indices;
4. top-30 feature names;
5. patient-specific decision threshold;
6. training-set size and class counts;
7. metadata describing the feature-extraction and model configuration.

These artifacts make the current project a suitable base for the next-stage software or firmware implementation.

Appendix C. Clinical Error Analysis and Refinement Branch

C.1 Branch Baseline and Interpretation

The v2-v5 notebooks used a reconciled clinical-analysis baseline of macro sensitivity 0.9778, macro FAR/h 0.3907, and mean delay 9.96 s across 54 seizures and 564.56 h. These values were stable across the later reruns.

The older headline benchmark in Appendix B comes from a different notebook snapshot and reports 0.98 sensitivity, 0.2455 FAR/h, 10.64 s mean delay, 55 seizures, and 580.57 h. Because the two outputs come from different evaluation paths, the later refinement results should be interpreted relative to the clinical-analysis baseline only.

C.2 Guard-Based Summary of Representative Variants

Table 12. Guard-based summary of representative variants from the clinical refinement branch

Variant	Macro Sens.	FAR/h	Mean Delay (s)	Key Benefit	Status
Clinical branch baseline	0.9778	0.3907	9.96	Reference branch used for V2-V5	Reference only
proxy_augmented_rf	0.9778	0.2575	9.42	Best global branch improvement	Accepted in branch
artifact_aware_rf	0.7873	0.1854	8.45	Large FAR reduction	Rejected: sensitivity collapse
proxy_hybrid_rf	0.9528	0.2551	9.63	Lower FAR/h than V2	Rejected: one chb04 miss
proxy_hybrid_plus_count_rf	0.9667	0.2810	11.07	Restored chb04 sensitivity	Rejected: later chb04 alarm
proxy_hybrid_plus_count_delayfirst_d2_rf	0.9778	1.2423	6.03	Earlier alarms	Rejected: false alarms exploded

C.3 Leakage-Control Notes

1. Top-k feature selection remains restricted to inner-train data.
2. Decision thresholds remain selected from inner-validation data.
3. Later cutoffs and refinement rules are derived from training-side statistics rather than from held-out test labels.
4. The larger methodological risk is repeated focus-driven iteration, not direct leakage. This is why the global adoption guard is retained in the refinement branch.